ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** photosynthesis	Using sunlight.
2	**(c)** transpiration	Evaporation.
3	**(b)** green	Colour.
4	**(b)** chlorophyll	Pigment.
5	**(b)** insects	Insectivorous.
6	**(c)** day	For CO2.
7	**(d)** stomata	Pores.
8	**(d)** parasite	Harms host.
9	**(b)** wastes	Excretion.
10	**(d)** roots	Root hairs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Food making by plants (1); Needs sunlight (1).	1+1
12	Small pores in leaves (1); Gas exchange (1).	1+1
13	Lives on and feeds on host (1); Cuscuta (1).	1+1
14	Chlorophyll (1); Green (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Carbon dioxide + water (1); Sunlight and chlorophyll (1); Glucose + oxygen (1).	1+1+1
16	Anchor plant (1); Absorb water (1); Absorb minerals (1).	1+1+1
17	Boil in alcohol (1); Add iodine (1); Blue-black colour (1).	1+1+1
	OR: Blade (1); Veins (1); Stalk (1).	1+1+1
18	Support (1); Carry water (1); Carry food (1).	1+1+1
	OR: Fix nitrogen (1); Enrich soil (1); Better crops (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Occurs in leaves (1); Needs light (1); Needs CO2 and water (1); Makes glucose (1); Gives oxygen (1).	1+1+1+1+1
	OR: Trap insects (1); Pitcher plant (1); Venus flytrap (1); Need nitrogen (1); Have green leaves (1).	1+1+1+1+1
20	Boil in water (1); Boil in alcohol (1); Wash (1); Add iodine (1); Note colour (1).	1+1+1+1+1
	OR: Bacteria (1); In root nodules (1); Fix nitrogen (1); Legumes (1); Symbiosis (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that photosynthesis takes place only at night, or that respiration takes place only at night; photosynthesis needs **light**, respiration goes on **all the time**.
- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** saprotroph	Dead matter.
2	**(b)** day and night	Always.
3	**(c)** sunlight	Light.
4	**(b)** fungus	Together.
5	**(d)** chloroplasts	Cell part.
6	**(b)** photosynthesis	By-product.
7	**(b)** pulses	Legume nodules.
8	**(b)** pale	No food.
9	**(d)** starch	Stored.
10	**(c)** nitrogen	Soil nutrient.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Loss of water vapour (1); From leaves (1).	1+1
12	Feeds on dead matter (1); Fungi (1).	1+1
13	Carbon dioxide (1); Water (1).	1+1
14	Xylem (1); Upward (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Autotrophic (1); Parasitic (1); Saprotrophic (1).	1+1+1
16	Sunlight (1); Water (1); Carbon dioxide (1).	1+1+1
17	Through stomata (1); By transpiration (1); As vapour (1).	1+1+1
	OR: Mango (1); Rose (1); Neem (1).	1+1+1
18	Glucose (1); Starch (1); Oxygen (1).	1+1+1
	OR: Remove wastes (1); Save water (1); In autumn (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tiny pores (1); On leaves (1); Guard cells (1); Gas exchange (1); Water vapour out (1).	1+1+1+1+1
	OR: Food (1); Oxygen (1); Wood (1); Medicines (1); Fibre (1).	1+1+1+1+1
20	Green pigment (1); In chloroplasts (1); Traps light (1); Needed for photosynthesis (1); Absent in fungi (1).	1+1+1+1+1
	OR: Water loss (1); Through stomata (1); Cools plant (1); Pulls water (1); Faster in the sun (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that photosynthesis takes place only at night, or that respiration takes place only at night; photosynthesis needs **light**, respiration goes on **all the time**.
- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** energy	Light.
2	**(c)** starch	Blue-black.
3	**(b)** producers	Food makers.
4	**(c)** plants	For food.
5	**(d)** xylem	Vessels.
6	**(a)** food	Food.
7	**(d)** blue-black	Positive test.
8	**(a)** phloem	Vessels.
9	**(a)** oxygen	Breathable air.
10	**(c)** soil	Anchorage.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pitcher plant (1); Venus flytrap (1).	1+1
12	Oxygen (1); By-product (1).	1+1
13	Phloem (1); To all parts (1).	1+1
14	Rhizobium (1); Fixes nitrogen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cuscuta (1); Mushroom (1); Yeast (1).	1+1+1
16	Chlorophyll (1); Absorbs light (1); Makes food (1).	1+1+1
17	Makes food (1); Gives oxygen (1); Uses CO_2 (1).	1+1+1
	OR: Make food (1); Release oxygen (1); Transpiration (1).	1+1+1
18	For energy (1); For making food (1); For growth (1).	1+1+1
	OR: Light (1); Heat (1); Energy (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Xylem water (1); Phloem food (1); Xylem upward (1); Phloem both directions (1); Different cells (1).	1+1+1+1+1
	OR: Leaves are kitchen (1); Water from root (1); Gas from air (1); Light from sun (1); Food stored (1).	1+1+1+1+1
20	Take food from host (1); Cuscuta (1); Yellow (1); No chlorophyll (1); Harm the host (1).	1+1+1+1+1
	OR: Food (1); Oxygen (1); Shelter (1); Medicine (1); Clean air (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that photosynthesis takes place only at night, or that respiration takes place only at night; photosynthesis needs **light**, respiration goes on **all the time**.
- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.